

使用 RISC-V

任务 1：斐波那契数列（6 分）

请编写汇编程序，计算斐波那契数列的第 n 项（ $n \leq 20$ ）。初始时， n 的值保存在 $a0$ 中。程序执行完成后，数列的第 n 项保存在 $a0$ 中。

$$F(0) = 0 \quad F(1) = 1 \quad F(n) = F(n-1) + F(n-2) \quad (n \geq 2)$$

任务 2：大整数处理（3 分）

请编写汇编程序，计算斐波那契数列的第 n 项（ $n \leq 45$ ）。初始时， n 的值保存在 $a0$ 中。程序执行完成后，数列的第 n 项保存在 $a0$ 和 $a1$ 中，其中 $a1$ 存储结果的高 32 位， $a0$ 存储结果的低 32 位。

任务 3：导出 COE 文件（1 分）

完成汇编程序的编写之后，请导出指令段的 Coefficient File 文件，以供后续实验使用。

COE 文件

COE 文件是 Vivado 存储器 IP 核的初始化文件。该文件可以将数据自动导入存储器中，以供其他模块单元的使用。它的一般格式如下：

```
memory_initialization_radix = 16;  
memory_initialization_vector =  
00008737  
f0070713  
000087b7  
f0478793
```

其中，`memory_initialization_radix` 指示数据使用的进制，`memory_initialization_vector` 则给出了每一个存储单元保存的数据。在上面的例子里，我们有

$M[0] = 00008737$ $M[1] = f0070713$ $M[2] = 000087b7$ $M[3] = f0478793$

对于其他存储单元，初始时将被置为 0。

Report

Task 1

How to run the code in Rars?

Note: I am not supposed to store function variables in the stack pointer. It is just very strange and unconventional to do so. So I stored the registers in $a0$.

Note: The first task is included in the second task.

Just move the $a0$ to sp and the output to gp if you want.

- 1. 在 rars 里新建一个文件
- 2. 粘贴代码（或者编辑代码）
- 3. Press Run>Assemble

Registers	Floating Point	Control and Status	
Name	Number	Value	
zero	0	0x00000000	
ra	1	0x00000000	
sp	2	0x7ffffc	
gp	3	0x10008000	
tp	4	0x00000000	
t0	5	0x00000000	
t1	6	0x00000000	
t2	7	0x00000000	
s0	8	0x00000000	
s1	9	0x00000000	
a0	10	0x00000001	
a1	11	0x00000000	
a2	12	0x00000000	
a3	13	0x00000000	
a4	14	0x00000000	
a5	15	0x00000000	
a6	16	0x00000000	
a7	17	0x00000000	
s2	18	0x00000000	
s3	19	0x00000000	
s4	20	0x00000000	
s5	21	0x00000000	
s6	22	0x00000000	
s7	23	0x00000000	
s8	24	0x00000000	
s9	25	0x00000000	
s10	26	0x00000000	
s11	27	0x00000000	
t3	28	0x00000000	
t4	29	0x00000000	
t5	30	0x00000000	
t6	31	0x00000000	
pc		0x00400000	

- 4. Change the Vars
- 5. Run>Reset
- 6. Run>Go

Debugging: Run>Step

If input is 1:

Registers	Floating Point	Control and Status	
Name	Number	Value	
zero	0	0x00000000	
ra	1	0x00000000	
sp	2	0x7ffffeffc	
gp	3	0x10008000	
tp	4	0x00000000	
t0	5	0x00000000	
t1	6	0x00000000	
t2	7	0x00000000	
s0	8	0x00000000	
s1	9	0x00000000	
a0	10	0x00000001	
a1	11	0x00000000	
a2	12	0x00000000	
a3	13	0x00000000	
a4	14	0x00000000	
a5	15	0x00000000	
a6	16	0x00000000	
a7	17	0x00000000	
s2	18	0x00000000	
s3	19	0x00000000	
s4	20	0x00000000	
s5	21	0x00000000	
s6	22	0x00000000	
s7	23	0x00000000	
s8	24	0x00000000	
s9	25	0x00000000	
s10	26	0x00000000	
s11	27	0x00000000	
t3	28	0x00000000	
t4	29	0x00000000	
t5	30	0x00000000	
t6	31	0x00000000	
pc		0x00400000	

Registers	Floating Point	Control and Status	
Name	Number	Value	
zero	0	0x00000000	
ra	1	0x00000000	
sp	2	0x7ffffeffc	
gp	3	0x10008000	
tp	4	0x00000000	
t0	5	0x00000001	
t1	6	0x00000000	
t2	7	0x00000000	
s0	8	0x00000000	
s1	9	0x00000000	
a0	10	0x00000001	
a1	11	0x00000000	
a2	12	0x00000000	
a3	13	0x00000000	
a4	14	0x00000000	
a5	15	0x00000000	
a6	16	0x00000000	
a7	17	0x00000000	
s2	18	0x00000000	
s3	19	0x00000000	
s4	20	0x00000000	
s5	21	0x00000000	
s6	22	0x00000000	
s7	23	0x00000000	
s8	24	0x00000000	
s9	25	0x00000000	
s10	26	0x00000000	
s11	27	0x00000000	
t3	28	0x00000000	
t4	29	0x00000000	
t5	30	0x00000000	
t6	31	0x00000000	
pc		0x00000004	

If input is 2:

Registers	Floating Point	Control and Status	
Name	Number	Value	
zero	0	0x00000000	
ra	1	0x00000000	
sp	2	0x7ffefffc	
gp	3	0x10008000	
tp	4	0x00000000	
t0	5	0x00000001	
t1	6	0x00000001	
t2	7	0x00000001	
s0	8	0x00000000	
s1	9	0x00000000	
a0	10	0x00000001	
a1	11	0x00000000	
a2	12	0x00000000	
a3	13	0x00000000	
a4	14	0x00000000	
a5	15	0x00000000	
a6	16	0x00000000	
a7	17	0x00000000	
s2	18	0x00000000	
s3	19	0x00000000	
s4	20	0x00000000	
s5	21	0x00000000	
s6	22	0x00000000	
s7	23	0x00000000	
s8	24	0x00000000	
s9	25	0x00000000	
s10	26	0x00000000	
s11	27	0x00000000	
t3	28	0x00000000	
t4	29	0x00000000	
t5	30	0x00000000	
t6	31	0x00000000	
pc		0x00000004	

If input is 10(a in base 16)

Registers	Floating Point	Control and Status	
Name	Number	Value	
zero	0	0x00000000	
ra	1	0x00000000	
sp	2	0x7ffffeffc	
gp	3	0x10008000	
tp	4	0x00000000	
t0	5	0x00000000	
t1	6	0x00000000	
t2	7	0x00000000	
s0	8	0x00000000	
s1	9	0x00000000	
a0	10	0x0000000a	
a1	11	0x00000000	
a2	12	0x00000000	
a3	13	0x00000000	
a4	14	0x00000000	
a5	15	0x00000000	
a6	16	0x00000000	
a7	17	0x00000000	
s2	18	0x00000000	
s3	19	0x00000000	
s4	20	0x00000000	
s5	21	0x00000000	
s6	22	0x00000000	
s7	23	0x00000000	
s8	24	0x00000000	
s9	25	0x00000000	
s10	26	0x00000000	
s11	27	0x00000000	
t3	28	0x00000000	
t4	29	0x00000000	
t5	30	0x00000000	
t6	31	0x00000000	
pc		0x00400000	

Registers	Floating Point	Control and Status	
Name	Number	Value	
zero	0	0x00000000	
ra	1	0x00000000	
sp	2	0x7ffffefc	
gp	3	0x10008000	
tp	4	0x00000000	
t0	5	0x00000022	
t1	6	0x00000037	
t2	7	0x00000037	
s0	8	0x00000000	
s1	9	0x00000000	
a0	10	0x00000037	
a1	11	0x00000000	
a2	12	0x00000000	
a3	13	0x00000000	
a4	14	0x00000000	
a5	15	0x00000000	
a6	16	0x00000000	
a7	17	0x00000000	
s2	18	0x00000000	
s3	19	0x00000000	
s4	20	0x00000000	
s5	21	0x00000000	
s6	22	0x00000000	
s7	23	0x00000000	
s8	24	0x00000000	
s9	25	0x00000000	
s10	26	0x00000000	
s11	27	0x00000000	
t3	28	0x00000000	
t4	29	0x00000000	
t5	30	0x00000000	
t6	31	0x00000000	
pc		0x00000004	

Task 2

First test the input being 1, 2, 0xa in base 16 and everything works.

Set input to 20(0x14) and output 0x1a6d=6765

Registers	Floating Point	Control and Status	
Name	Number	Value	
zero	0	0x00000000	
ra	1	0x00000000	
sp	2	0x7ffefffc	
gp	3	0x10008000	
tp	4	0x00000000	
t0	5	0x00000000	
t1	6	0x00000000	
t2	7	0x00000000	
s0	8	0x00000000	
s1	9	0x00000000	
a0	10	0x00000014	
a1	11	0x00000000	
a2	12	0x00000000	
a3	13	0x00000000	
a4	14	0x00000000	
a5	15	0x00000000	
a6	16	0x00000000	
a7	17	0x00000000	
s2	18	0x00000000	
s3	19	0x00000000	
s4	20	0x00000000	
s5	21	0x00000000	
s6	22	0x00000000	
s7	23	0x00000000	
s8	24	0x00000000	
s9	25	0x00000000	
s10	26	0x00000000	
s11	27	0x00000000	
t3	28	0x00000000	
t4	29	0x00000000	
t5	30	0x00000000	
t6	31	0x00000000	
pc		0x00400000	

Registers	Floating Point	Control and Status	
Name	Number	Value	
zero	0	0x00000000	
ra	1	0x00000000	
sp	2	0x7ffefffc	
gp	3	0x10008000	
tp	4	0x00000000	
t0	5	0x00001055	
t1	6	0x00000000	
t2	7	0x00001a6d	
s0	8	0x00000000	
s1	9	0x00000000	
a0	10	0x00001a6d	
a1	11	0x00000000	
a2	12	0x00000000	
a3	13	0x00000000	
a4	14	0x00000000	
a5	15	0x00000000	
a6	16	0x00000000	
a7	17	0x00000000	
s2	18	0x00000000	
s3	19	0x00000000	
s4	20	0x00000000	
s5	21	0x00000000	
s6	22	0x00000000	
s7	23	0x00000000	
s8	24	0x00000000	
s9	25	0x00000000	
s10	26	0x00000000	
s11	27	0x00000000	
t3	28	0x00000000	
t4	29	0x00001a6d	
t5	30	0x00000000	
t6	31	0x00000000	
pc		0x00000004	

Set input to 30(0x1e) and output 0xcb228=832040

Registers	Floating Point	Control and Status	
Name	Number	Value	
zero	0	0x00000000	
ra	1	0x00000000	
sp	2	0x7ffefffc	
gp	3	0x10008000	
tp	4	0x00000000	
t0	5	0x00000000	
t1	6	0x00000000	
t2	7	0x00000000	
s0	8	0x00000000	
s1	9	0x00000000	
a0	10	0x0000001e	
a1	11	0x00000000	
a2	12	0x00000000	
a3	13	0x00000000	
a4	14	0x00000000	
a5	15	0x00000000	
a6	16	0x00000000	
a7	17	0x00000000	
s2	18	0x00000000	
s3	19	0x00000000	
s4	20	0x00000000	
s5	21	0x00000000	
s6	22	0x00000000	
s7	23	0x00000000	
s8	24	0x00000000	
s9	25	0x00000000	
s10	26	0x00000000	
s11	27	0x00000000	
t3	28	0x00000000	
t4	29	0x00000000	
t5	30	0x00000000	
t6	31	0x00000000	
pc		0x00400000	

Registers	Floating Point	Control and Status	
Name	Number	Value	
zero	0	0x00000000	
ra	1	0x00000000	
sp	2	0x7ffefffc	
gp	3	0x10008000	
tp	4	0x00000000	
t0	5	0x0007d8b5	
t1	6	0x00000000	
t2	7	0x000cb228	
s0	8	0x00000000	
s1	9	0x00000000	
a0	10	0x000cb228	
a1	11	0x00000000	
a2	12	0x00000000	
a3	13	0x00000000	
a4	14	0x00000000	
a5	15	0x00000000	
a6	16	0x00000000	
a7	17	0x00000000	
s2	18	0x00000000	
s3	19	0x00000000	
s4	20	0x00000000	
s5	21	0x00000000	
s6	22	0x00000000	
s7	23	0x00000000	
s8	24	0x00000000	
s9	25	0x00000000	
s10	26	0x00000000	
s11	27	0x00000000	
t3	28	0x00000000	
t4	29	0x000cb228	
t5	30	0x00000000	
t6	31	0x00000000	
pc		0x00000004	

Set input to 50(0x32) and output 0x2 and 0xee333961=12586269025

Registers	Floating Point	Control and Status	
Name	Number	Value	
zero	0	0x00000000	
ra	1	0x00000000	
sp	2	0x7ffefffc	
gp	3	0x10008000	
tp	4	0x00000000	
t0	5	0x00000000	
t1	6	0x00000000	
t2	7	0x00000000	
s0	8	0x00000000	
s1	9	0x00000000	
a0	10	0x00000032	
a1	11	0x00000000	
a2	12	0x00000000	
a3	13	0x00000000	
a4	14	0x00000000	
a5	15	0x00000000	
a6	16	0x00000000	
a7	17	0x00000000	
s2	18	0x00000000	
s3	19	0x00000000	
s4	20	0x00000000	
s5	21	0x00000000	
s6	22	0x00000000	
s7	23	0x00000000	
s8	24	0x00000000	
s9	25	0x00000000	
s10	26	0x00000000	
s11	27	0x00000000	
t3	28	0x00000000	
t4	29	0x00000000	
t5	30	0x00000000	
t6	31	0x00000000	
pc		0x00400000	

Registers	Floating Point	Control and Status	
Name	Number	Value	
zero	0	0x00000000	
ra	1	0x00000000	
sp	2	0x7ffefffc	
gp	3	0x10008000	
tp	4	0x00000000	
t0	5	0xcfa62f21	
t1	6	0x00000001	
t2	7	0xee333961	
s0	8	0x00000000	
s1	9	0x00000000	
a0	10	0xee333961	
a1	11	0x00000002	
a2	12	0x00000000	
a3	13	0x00000000	
a4	14	0x00000000	
a5	15	0x00000000	
a6	16	0x00000000	
a7	17	0x00000000	
s2	18	0x00000000	
s3	19	0x00000000	
s4	20	0x00000000	
s5	21	0x00000000	
s6	22	0x00000000	
s7	23	0x00000000	
s8	24	0x00000000	
s9	25	0x00000000	
s10	26	0x00000000	
s11	27	0x00000000	
t3	28	0x00000002	
t4	29	0xee333961	
t5	30	0x00000002	
t6	31	0x00000000	
pc		0x00000004	

Task 3

Mar 19 21:02

/home/user/Downloads/riscv3.asm - RARS 1.6

File Edit Run Settings Tools Help

Run speed at max (no interaction)

Text Segment

Bkpt	Address	Code	Basic	Source
☒	0x00400028	0x007ebfb3 sltu x31, x29, x7	22: sltu t6, t4, t2	# t5 = (t4 < t2) ...
☒	0x0040002c	0x01c3bf33 add x30, x6, x28	23: add t5, t1, t3	# t5 = t1 + t3 +
☒	0x00400030	0x01fffb33 add x30, x28, x31	24: add t5, t5, t6	#
☒	0x00400034	0x007002b3 add x5, x0, x7	27: mv t0, t2	# F(n-2) = F(n-1)
☒	0x00400038	0x01c00333 add x6, x0, x28	28: mv t1, t3	# F(n-2) = F(n-1)
☒	0x0040003c	0x01d003b3 add x7, x0, x29	29: mv t2, t4	# F(n-1) =
☒	0x00400040	0x01d003b3 add x28, x0, x30	30: mv t3, t5	# F(n-1) =
☒	0x00400044	0xff5f5813 addi x10, x10, 0xffffffffff	32: addi a0, a0, -1	# n = n - 1
☒	0x00400048	0xf09ff06f jal x0, 0xffffffffff	33: j loop	#
☒	0x0040004c	0x00700533 add x10, x0, x7	36: mv a0, t2	# 32 a0
☒	0x00400050	0x01c005b3 add x11, x0, x28	37: mv a1, t3	# 32 a1
☒	0x00400054	0x00000067 jalr x0, x1, 0	38: ret	#
☒	0x00400058	0x00000513 addi x10, x0, 0	41: li a0, 0	# 32 = 0
☒	0x0040005c	0x00000593 addi x11, x0, 0	42: li a1, 0	# 32 = 0
☒	0x00400060	0x00000067 jalr x0, x1, 0	43: ret	#
☒	0x00400064	0x00100513 addi x10, x0, 1	46: li a0, 1	# 32 = 1
☒	0x00400068	0x00000593 addi x11, x0, 0	47: li a1, 0	# 32 = 0
☒	0x0040006c	0x00000067 jalr x0, x1, 0	48: ret	#

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+C)
0x10010000	0x00000000	0x00000000	0x00000000	0x00
0x10010020	0x00000000	0x00000000	0x00000000	0x00
0x10010040	0x00000000	0x00000000	0x00000000	0x00
0x10010060	0x00000000	0x00000000	0x00000000	0x00
0x10010080	0x00000000	0x00000000	0x00000000	0x00
0x100100a0	0x00000000	0x00000000	0x00000000	0x00
0x100100c0	0x00000000	0x00000000	0x00000000	0x00
0x100100e0	0x00000000	0x00000000	0x00000000	0x00
0x10010100	0x00000000	0x00000000	0x00000000	0x00
0x10010120	0x00000000	0x00000000	0x00000000	0x00
0x10010140	0x00000000	0x00000000	0x00000000	0x00
0x10010160	0x00000000	0x00000000	0x00000000	0x00
0x10010180	0x00000000	0x00000000	0x00000000	0x00
0x100101a0	0x00000000	0x00000000	0x00000000	0x00
0x100101c0	0x00000000	0x00000000	0x00000000	0x00
0x100101e0	0x00000000	0x00000000	0x00000000	0x00

Dump Memory To File

Memory Segment: text (0x00400000 - 0x0040006c)

Dump Format: Text/Data Segment Window

Dump To File... Cancel

Registers

Name	Number	Value
zero	0	0x00000000
ra	1	0x00000000
sp	2	0x7ffffcfc
gp	3	0x10000000
tp	4	0x00000000
t0	5	0x00000000
t1	6	0x00000000
t2	7	0x00000000
s0	8	0x00000000
s1	9	0x00000000
a0	10	0x00000000
a1	11	0x00000000
a2	12	0x00000000
a3	13	0x00000000
a4	14	0x00000000
a5	15	0x00000000
a6	16	0x00000000
a7	17	0x00000000
s2	18	0x00000000
s3	19	0x00000000
s4	20	0x00000000
s5	21	0x00000000
s6	22	0x00000000
s7	23	0x00000000
s8	24	0x00000000
s9	25	0x00000000
s10	26	0x00000000
s11	27	0x00000000
t3	28	0x00000000
t4	29	0x00000000
t5	30	0x00000000
t6	31	0x00000000
pc		0x00400000

Messages

Run I/O

Go: execution terminated with errors.

Go: running riscv3.asm

Error in : Instruction load access error

Go: execution terminated with errors.

Clear

Assemble: assembling /home/user/Downloads/riscv3.asm

Assemble: operation completed successfully.

Dump COE:

1. Run>Assemble
2. File>Dump memory

Mar 19 22:07

/home/user/Downloads/riscv1.asm - RARS 1.6

File Edit Run Settings Tools Help

Run speed at max (no interaction)

Text Segment

Bkpt	Address	Code	Basic	Source
☒	0x00400000	0x04050c63 beq x10, x0, 0x00000058	3: beqz a0, return_0	# n = 0, 0
☒	0x00400004	0x00100293 addi x5, x0, 1	4: li t0, 1	#
☒	0x00400008	0x04550e63 beq x10, x5, 0x0000005c	5: beq a0, t0, return_1	# n = 1, 1
☒	0x0040000c	0x00000293 addi x5, x0, 0	8: li t0, 0	# F(0) 32
☒	0x00400010	0x00000313 addi x6, x0, 0	9: li t1, 0	# F(0) 32
☒	0x00400014	0x00100393 addi x7, x0, 1	10: li t2, 1	# F(1) 32
☒	0x00400018	0x00000e13 addi x28, x0, 0	11: li t3, 0	# F(1) 32
☒	0x0040001c	0xff5f5813 addi x10, x10, 0xffffffffff	13: addi a0, a0, -2	# n = n - 2,
☒	0x00400020	0x02054663 blt x10, x0, 0x0000002c	16: bltz a0, end_loop	# a0 < 0,
☒	0x00400024	0x00720eb3 add x29, x5, x7	20: add t4, t0, t2	# t4 = t0 + t2 ()
☒	0x00400028	0x007ebfb3 sltu x31, x29, x7	22: sltu t6, t4, t2	# t5 = (t4 < t2) ...
☒	0x0040002c	0x01c3bf33 add x30, x6, x28	23: add t5, t1, t3	# t5 = t1 + t3 +
☒	0x00400030	0x01fffb33 add x30, x28, x31	24: add t5, t5, t6	#
☒	0x00400034	0x007002b3 add x5, x0, x7	27: mv t0, t2	# F(n-2) = F(n-1)
☒	0x00400038	0x01c00333 add x6, x0, x28	28: mv t1, t3	# F(n-2) = F(n-1)
☒	0x0040003c	0x01d003b3 add x7, x0, x29	29: mv t2, t4	# F(n-1) =
☒	0x00400040	0x01d003b3 add x28, x0, x30	30: mv t3, t5	# F(n-1) =
☒	0x00400044	0xff5f5813 addi x10, x10, 0xffffffffff	32: addi a0, a0, -1	# n = n - 1

Data Segment

Address	Value (+0)	Value (+4)	Value (+8)	Value (+C)
0x10010000	0x00000000	0x00000000	0x00000000	0x00
0x10010020	0x00000000	0x00000000	0x00000000	0x00
0x10010040	0x00000000	0x00000000	0x00000000	0x00
0x10010060	0x00000000	0x00000000	0x00000000	0x00
0x10010080	0x00000000	0x00000000	0x00000000	0x00
0x100100a0	0x00000000	0x00000000	0x00000000	0x00
0x100100c0	0x00000000	0x00000000	0x00000000	0x00
0x100100e0	0x00000000	0x00000000	0x00000000	0x00
0x10010100	0x00000000	0x00000000	0x00000000	0x00
0x10010120	0x00000000	0x00000000	0x00000000	0x00
0x10010140	0x00000000	0x00000000	0x00000000	0x00
0x10010160	0x00000000	0x00000000	0x00000000	0x00
0x10010180	0x00000000	0x00000000	0x00000000	0x00
0x100101a0	0x00000000	0x00000000	0x00000000	0x00
0x100101c0	0x00000000	0x00000000	0x00000000	0x00
0x100101e0	0x00000000	0x00000000	0x00000000	0x00

Dump Memory To File

Memory Segment: text (0x00400000 - 0x0040006c)

Dump Format: Text/Data Segment Window

Dump To File... Cancel

Registers

Name	Number	Value
zero	0	0x00000000
ra	1	0x00000000
sp	2	0x7ffffcfc
gp	3	0x10000000
tp	4	0x00000000
t0	5	0x00000000
t1	6	0x00000000
t2	7	0x00000000
s0	8	0x00000000
s1	9	0x00000000
a0	10	0x00000000
a1	11	0x00000000
a2	12	0x00000000
a3	13	0x00000000
a4	14	0x00000000
a5	15	0x00000000
a6	16	0x00000000
a7	17	0x00000000
s2	18	0x00000000
s3	19	0x00000000
s4	20	0x00000000
s5	21	0x00000000
s6	22	0x00000000
s7	23	0x00000000
s8	24	0x00000000
s9	25	0x00000000
s10	26	0x00000000
s11	27	0x00000000
t3	28	0x00000000
t4	29	0x00000000
t5	30	0x00000000
t6	31	0x00000000
pc		0x00400000

Messages

Run I/O

Assemble: assembling /home/user/Downloads/riscv1.asm

Assemble: operation completed successfully.

Clear